

IBMP

Pain Medicine

Confidential answer key and marking guide / Set A

For faculty only. Do not distribute this file to candidates.

Assessment draft for trained medical learners. Review against the taught syllabus, local protocols and applicable law before release. No institutional approval, accreditation or validation is implied.

Assessment blueprint

Module	MCQs	Images	Short	Marks
M1. Foundations of Pain Medicine	7	1	2	12
M2. Pharmacologic Pain Management	7	1	2	12
M3. Interventional Pain Procedures - Spine	6	2	1	13
M4. Interventional Pain Procedures - Non-Spine	5	2	1	12
M5. Cancer Pain and Palliative Care	5	1	1	9
M6. Chronic Pain Syndromes and Rehabilitation	6	2	2	14
M7. Neuromodulation and Advanced Therapies	4	1	1	8
TOTAL	40	10	10	80

The module allocation is a proposed blueprint, not an IBMP policy. Viva and project marks remain outside this 80-mark paper.

Section A | MCQ key

Q	Key	Q	Key	Q	Key	Q	Key
Q01	C	Q11	D	Q21	A	Q31	C
Q02	A	Q12	B	Q22	C	Q32	B
Q03	D	Q13	A	Q23	B	Q33	B
Q04	D	Q14	C	Q24	A	Q34	D
Q05	B	Q15	B	Q25	C	Q35	A
Q06	A	Q16	D	Q26	A	Q36	D
Q07	B	Q17	A	Q27	C	Q37	C
Q08	A	Q18	D	Q28	D	Q38	A
Q09	B	Q19	B	Q29	D	Q39	C
Q10	C	Q20	C	Q30	B	Q40	D

Marking rules

- MCQs: 1 mark for the keyed response; 0 for an incorrect, multiple or missing response. Negative marking is not proposed.
- Image questions: 3 marks each, normally 1 for each subpart. Q45(b) explicitly allows 0.5 + 0.5 marks. Accept clinically equivalent phrasing.
- Short answers: maximum 1 mark each. Accept one valid answer where one is requested. Do not require an essay for a one-mark item.
- Score subparts independently unless the rubric explicitly requires a link. Do not double-penalise a single error. Review contradictory or potentially unsafe responses clinically.
- Publish results only after manual marking and moderation. Confirm duration, pass standard, accommodations, appeals and retest rules before use.

Faculty checks before release

- f) Check all 60 stems and distractors against the teaching delivered; pilot for difficulty, timing and item discrimination.
- g) Have a pain physician and relevant imaging/procedural faculty approve the figures. They are original schematic stimuli, not validated diagnostic images.
- h) If actual scan interpretation is a course outcome, replace relevant schematics with licensed, de-identified, faculty-validated clinical images and re-check the answer key.
- i) Where guidance varies, the named guideline defines the intended answer. Do not apply the CDC chronic-pain opioid recommendations to cancer or palliative populations.
- j) For final delivery, keep question and figure together. Randomise MCQ option order only if the scoring key remains linked to answer content. Keep image subparts grouped.
- k) Import MCQs as single-best-answer items. Use manually marked written-response items for image subparts and the short answers; set section totals to 40, 30 and 10.

MCQ rationales and curriculum mapping

Q01 | C | M1 - Pain physiology

Thinly myelinated A-delta fibres. A-delta fibres carry fast, localised nociceptive signals; C fibres carry slower signals.

Reference basis: S02

Q02 | A | M1 - Neurobiology

Central sensitisation. The defining change is increased responsiveness within the central nervous system.

Reference basis: S01

Q03 | D | M1 - Pain classification

Allodynia. Allodynia refers to pain provoked by ordinarily non-painful stimulation.

Reference basis: S01

Q04 | D | M1 - Assessment tools

Interference with sleep, activity and participation. Intensity should be supplemented by functional and psychosocial assessment.

Reference basis: S03

Q05 | B | M1 - Psychosocial aspects

Explore the belief and agree a paced, graded return to meaningful activity. A collaborative plan can address fear and support safe participation.

Reference basis: S03

Q06 | A | M1 - Ethics and consent

Respect the decision, document it and discuss acceptable alternatives. A capacitated patient may decline an elective intervention; consent is an informed dialogue.

Reference basis: S04

Q07 | B | M1 - Communication

Ask the patient to explain the plan and key precautions in their own words. Teach-back checks understanding through the patient's own explanation.

Reference basis: S05

Q08 | A | M2 - Non-opioid analgesics

Further renal impairment from reduced renal prostaglandin activity. Renal prostaglandin inhibition can compromise kidney function in susceptible patients.

Reference basis: S06

Q09 | B | M2 - Paracetamol safety

Unrecognised duplication can produce a hepatotoxic total dose. All paracetamol-containing products contribute to cumulative exposure and overdose risk.

Reference basis: S07

Q10 | C | M2 - Adjuvant pharmacology

Renal function. Gabapentin is renally cleared, so impaired renal function requires dose adjustment.

Reference basis: S08

Q11 | D | M2 - Neuropathic pain treatment

Duloxetine. Duloxetine is one guideline-supported initial option for this neuropathic pain context.

Reference basis: S09

Q12 | B | M2 - Opioid interactions

Sedation and respiratory depression. Concurrent central nervous system depressants increase overdose and respiratory risk.

Reference basis: S10

Q13 | A | M2 - Opioid stewardship

Agree an individualised, supported reduction plan and optimise other care. Absent an urgent safety threat, avoid abrupt cessation and use a collaborative taper.

Reference basis: S10

Q14 | C | M2 - Risk mitigation

Agree measurable goals, assess risks, explain precautions and arrange review. Opioid care requires goals, risk assessment and ongoing benefit-harm review.

Reference basis: S10

Q15 | B | M3 - Epidural steroid injections

Acute, severe sciatica after clinical assessment. NICE identifies acute severe sciatica as a context in which an epidural may be considered.

Reference basis: S11

Q16 | D | M3 - Medial branch blocks

A clinically meaningful response to a properly performed diagnostic medial branch block. Diagnostic medial branch response is used to inform selection; imaging alone is insufficient.

Reference basis: S11, S12

Q17 | A | M3 - Radiofrequency ablation

Medial branches carrying sensation from the facet joints. Lumbar facet radiofrequency targets the relevant medial branch innervation.

Reference basis: S12

Q18 | D | M3 - Discogenic pain interventions

Correlate symptoms and examination, review alternatives and explain procedure-specific evidence. Degenerative imaging does not by itself establish the source of pain or justify an invasive treatment.

Reference basis: S11, S13

Q19 | B | M3 - Image-guided spine safety

Image guidance assists targeting but does not eliminate serious complications. Serious neurological events have been reported even with fluoroscopic guidance.

Reference basis: S14

Q20 | C | M3 - Post-procedure complication

Arrange immediate emergency assessment for a possible compressive neurological complication. New progressive neurological or sphincter symptoms require urgent assessment, not routine follow-up.

Reference basis: S13

Q21 | A | M4 - Peripheral nerve blocks

Inhibition of voltage-gated sodium channels. Sodium-channel inhibition interrupts propagation of nerve impulses.

Reference basis: S15

Q22 | C | M4 - Joint injection safety

Defer the steroid injection and urgently assess for joint infection. Possible septic arthritis requires urgent investigation; therapeutic steroid injection should be deferred.

Reference basis: S16

Q23 | B | M4 - Sympathetic blocks

Celiac plexus block or neurolysis. The celiac plexus carries visceral pain pathways from upper abdominal organs.

Reference basis: S17

Q24 | A | M4 - Myofascial pain

A tender spot within a taut muscle band that reproduces the patient's familiar pain. Reproduction of familiar local or referred pain from a tender taut band supports the diagnosis.

Reference basis: S18

Q25 | C | M4 - Ultrasound guidance

Identify relevant vessels and maintain awareness of the needle tip throughout advancement. Ultrasound helps identify anatomy; tip localisation and vascular awareness remain essential.

Reference basis: S19

Q26 | A | M5 - Cancer pain assessment

Physical symptoms together with psychological, social and spiritual concerns. Comprehensive cancer pain care addresses multiple interacting dimensions of distress.

Reference basis: S20

Q27 | C | M5 - WHO analgesic approach

Select analgesia appropriate to severity, including an opioid when indicated, with reassessment. Treatment should match pain severity; severe pain need not pass through a prolonged mild-analgesic-only stage.

Reference basis: S21

Q28 | D | M5 - Intrathecal drug delivery

A patient with refractory cancer pain or intolerable systemic adverse effects despite optimised care. Selected refractory cases may benefit, but specialist assessment and follow-up are necessary.

Reference basis: S22

Q29 | D | M5 - End-of-life pain management

Proportionate symptom relief consistent with the patient's goals, with appropriate reassessment. The therapeutic aim is appropriate symptom relief, not an unexamined fixed regimen.

Reference basis: S20, S21

Q30 | B | M5 - Multidisciplinary palliative care

Alongside active cancer treatment when symptom or support needs warrant it. Palliative input can accompany disease-directed treatment and is based on needs.

Reference basis: S20

Q31 | C | M6 - Low back pain red flags

Urinary retention with saddle sensory change. Sphincter disturbance with saddle sensory change is a neurological red flag.

Reference basis: S13

Q32 | B | M6 - Neuropathic pain syndromes

Neuropathic pain. Postherpetic neuralgia involves injury or disease of the somatosensory system.

Reference basis: S01

Q33 | B | M6 - Complex regional pain syndrome

Use clinical criteria incorporating symptoms, examination signs and exclusion of a better explanation. CRPS is a clinical diagnosis; tests may help investigate alternatives but are not a stand-alone confirmation.

Reference basis: S23

Q34 | D | M6 - Fibromyalgia and centralised pain

Nociplastic pain. Nociplastic mechanisms can underlie widespread pain without a sufficient tissue or nerve-lesion explanation.

Reference basis: S01

Q35 | A | M6 - Rehabilitation

Progressively increasing an agreed meaningful activity while monitoring tolerability. Rehabilitation emphasises individualised participation and function, not inactivity until pain vanishes.

Reference basis: S03

Q36 | D | M6 - Trigeminal neuralgia

Carbamazepine. Trigeminal neuralgia has a distinct first-line recommendation: carbamazepine.

Reference basis: S09

Q37 | C | M7 - Spinal cord stimulation

Multidisciplinary selection, psychosocial assessment and an appropriate stimulation trial. Selection includes psychosocial assessment and usually a trial for these indications.

Reference basis: S24

Q38 | A | M7 - Intrathecal pumps

Seek urgent assessment by the responsible specialist or emergency service. Pump interruption or malfunction may cause withdrawal, loss of analgesia or other serious complications.

Reference basis: S25

Q39 | C | M7 - Regenerative medicine

Evaluate product-specific evidence, regulatory status, risks and uncertainty before counselling. Unproven regenerative claims require critical review; registration or marketing is not evidence of benefit.

Reference basis: S26

Q40 | D | M7 - Emerging therapies

Independent, well-conducted comparative trials reporting function, pain, harms and durability. Credible assessment needs relevant clinical outcomes, appropriate comparison and transparent harms reporting.

Reference basis: S26

Section B | Image marking guide

Each question carries 3 marks. Mark the three parts separately. Each figure appears in the corresponding student-paper question.

Q41 | Figure 1 | M1 - Pain assessment

- l) 7/10. [1 mark]
- m) One of: functional interference, sleep interference, mood/emotional impact, or participation/social impact. [1 mark]
- n) A relevant question, for example: "How does pain affect your walking or usual daily activities?"
Match the dimension in part (b). [1 mark]

Marker note: Do not require a universal mild/moderate/severe cut-off. Part (c) may earn credit independently if clinically relevant.

Reference basis: S03

Q42 | Figure 2 | M2 - Analgesic safety observation chart

- o) Opioid-induced respiratory depression / opioid toxicity with hypoventilation. [1 mark]
- p) Activate emergency help and promptly assess/support airway and ventilation; do not delay respiratory support while awaiting a medicine. [1 mark]
- q) Naloxone. [1 mark]

Marker note: For (b), award 1 mark for prompt airway/ventilation support with urgent escalation. Merely arranging a routine review is not sufficient.

Reference basis: S10, S27

Q43 | Figure 3 | M3 - Lumbar anatomy and clinical correlation

- r) Right paracentral/posterolateral L4-L5 disc herniation. [1 mark]
- s) Right L5 nerve root (the traversing root). [1 mark]
- t) New urinary retention/incontinence or new saddle/perineal sensory loss. [1 mark]

Marker note: Accept disc protrusion/herniation; no morphological subtype beyond the schematic is required. This is not a patient MRI.

Reference basis: S13

Q44 | Figure 4 | M3 - Posterior spinal pain structures

- u) Lumbar facet / zygapophyseal joint. [1 mark]
- v) Medial branches of the dorsal rami (posterior primary rami). [1 mark]
- w) Diagnostic lumbar medial branch block. [1 mark]

Marker note: Do not ask students to infer exact vertebral levels, needle positions or treatment parameters from this simplified figure.

Reference basis: S12

Q45 | Figure 5 | M4 - Ultrasound pattern recognition

- x) Peripheral nerve. [1 mark]
- y) Artery (0.5 mark) and colour or power Doppler (0.5 mark). [1 mark]
- z) Keep the needle tip located during advancement; stop advancing if its position is uncertain. [1 mark]

Marker note: The figure is a stylised ultrasound schematic, not a scan. Do not award the safety mark for visualising only the needle shaft.

Reference basis: S19

Q46 | Figure 6 | M4 - Joint assessment before injection

- aa) Knee joint effusion / suprapatellar effusion. [1 mark]
- bb) Septic arthritis / joint infection. [1 mark]
- cc) Urgent clinical assessment and diagnostic joint aspiration for synovial analysis and culture; defer steroid injection. [1 mark]

Marker note: An effusion is not specific for infection. For (c), urgent aspiration/infection assessment earns the mark; in systemic sepsis, treatment must not be delayed.

Reference basis: S16

Q47 | Figure 7 | M5 - Cancer pain pattern

- dd) Breakthrough cancer pain. [1 mark]
- ee) An individually prescribed rescue analgesic, commonly an appropriate immediate-release opioid, matched to the patient and existing regimen. [1 mark]
- ff) Reassess the underlying cause/triggers or adequacy of background analgesia, adherence, adverse effects, or overall regimen. [1 mark]

Marker note: Do not award a medicine-strategy mark for slow-release opioid alone as immediate rescue. No fixed rescue dose is required.

Reference basis: S21

Q48 | Figure 8 | M6 - Pain distribution

gg) Painful diabetic distal symmetric polyneuropathy / painful diabetic peripheral neuropathy. [1 mark]

hh) Symmetrical distal stocking distribution. [1 mark]

ii) One of: duloxetine, amitriptyline, gabapentin or pregabalin. [1 mark]

Marker note: Award credit for an appropriate equivalent description of a length-dependent distal symmetrical distribution.

Reference basis: S09, S28

Q49 | Figure 9 | M6 - Regional pain after injury

jj) Complex regional pain syndrome (CRPS); type I is acceptable if no major nerve injury is established. [1 mark]

kk) Vasomotor category. [1 mark]

ll) Graded functional restoration/physiotherapy or occupational therapy, with desensitisation and appropriate education. [1 mark]

Marker note: The stem supplies history and examination context. The drawing or a temperature difference alone does not diagnose CRPS.

Reference basis: S23

Q50 | Figure 10 | M7 - Implanted pain therapy

mm) Spinal cord stimulation (SCS). [1 mark]

nn) Epidural space. [1 mark]

oo) Psychosocial assessment (including mood), realistic expectations, functional goals, capacity to manage the device, or suitability/response during an appropriate trial. [1 mark]

Marker note: Accept one clinically valid selection domain. Do not infer a specific vertebral level or surgical technique.

Reference basis: S24, S29

Section C | Short-answer key

Q51 | M1 - Classification | 1 mark

Pain persisting or recurring for longer than 3 months; accept at least 3 months where used by the specified classification.

Reference basis: S30

Q52 | M1 - Consent | 1 mark

One procedure-relevant material risk, e.g. bleeding/haematoma, infection, nerve injury, dural puncture/headache or a serious drug reaction.

Reference basis: S04, S14

Q53 | M2 - Non-opioid pharmacology | 1 mark

Cyclo-oxygenase / COX (COX-1 and/or COX-2).

Reference basis: S06

Q54 | M2 - Opioid risk mitigation | 1 mark

One of: offer naloxone with education, review/avoid risky sedative combinations where feasible, assess overdose risks, or arrange closer monitoring and follow-up. Accept safe storage education if explicitly linked to preventing accidental overdose.

Reference basis: S10

Q55 | M3 - Procedure safety | 1 mark

Anticoagulants or antiplatelet medicines. A correct example such as warfarin, apixaban or clopidogrel is acceptable.

Reference basis: S31

Q56 | M4 - Local anaesthetic toxicity | 1 mark

Local anaesthetic systemic toxicity (LAST).

Reference basis: S15, S32

Q57 | M5 - Cancer pain | 1 mark

Palliative radiotherapy. Accept appropriate surgical stabilisation for a painful unstable/pathological fracture if specified.

Reference basis: S21

Q58 | M6 - Assessment of low back pain | 1 mark

Oswestry Disability Index (ODI) or Roland-Morris Disability Questionnaire (RMDQ).

Reference basis: S33

Q59 | M6 - Rehabilitation | 1 mark

Break activity into planned manageable periods with rests and progress gradually toward an agreed goal; avoid a repeated overactivity-and-crash pattern.

Reference basis: S03

Q60 | M7 - Neuromodulation selection | 1 mark

Depression/mood, anxiety, coping, expectations, or substance-use problems. Accept one relevant domain; screening should include depression.

Reference basis: S24

Reference notes

References support the clinical concepts and marking rationale; questions, fictional cases and figures were newly authored for this draft. Sources were checked on 3 September 2026. No third-party clinical images, rating-scale forms or assessment questionnaires are reproduced.

International guidance is used for academic context. Faculty must apply current local prescribing rules, product approvals, consent law and procedural protocols. The paper deliberately avoids fixed opioid-conversion doses, anticoagulant interruption intervals and procedural needle-placement instructions.

S01 IASP. Terminology: pain, sensitisation, allodynia, neuropathic and nociplastic pain. [Open source](#)

S02 StatPearls. Physiology, Pain. [Open source](#)

S03 NICE NG193. Chronic pain: assessment and management. Recommendations. [Open source](#)

S04 GMC. The seven principles of decision making and consent. General ethical principles; local law must be applied. [Open source](#)

S05 AHRQ. Tool: Teach-Back. [Open source](#)

S06 StatPearls. Nonsteroidal Anti-Inflammatory Drugs (NSAIDs). [Open source](#)

S07 FDA. Acetaminophen: avoiding duplicate products and overdose. [Open source](#)

S08 StatPearls. Gabapentin. [Open source](#)

S09 NICE CG173. Neuropathic pain in adults: recommendations. [Open source](#)

S10 CDC. Clinical Practice Guideline for Prescribing Opioids for Pain, 2022. Not applicable to cancer, sickle cell, palliative or end-of-life pain. [Open source](#)

S11 NICE NG59. Low back pain and sciatica: recommendations. [Open source](#)

S12 Cohen SP et al. Consensus practice guidelines on interventions for lumbar facet joint pain. Regional Anesthesia and Pain Medicine, 2020. [Open source](#)

S13 StatPearls. Lumbar Disc Herniation. [Open source](#)

S14 FDA. Safety communication: serious neurological problems after epidural corticosteroid injections. [Open source](#)

S15 StatPearls. Local Anesthetic Toxicity: local anaesthetic mechanisms and toxicity. [Open source](#)

- S16** American Family Physician. Joint and Soft Tissue Injection. Therapeutic injection versus diagnostic aspiration. [Open source](#)
- S17** StatPearls. Celiac Plexus Block. [Open source](#)
- S18** StatPearls. Myofascial Pain Syndrome. [Open source](#)
- S19** NYSORA. Introduction to Ultrasound-Guided Regional Anesthesia. [Open source](#)
- S20** WHO. Palliative care fact sheet. [Open source](#)
- S21** WHO. Guidelines for the pharmacological and radiotherapeutic management of cancer pain in adults and adolescents: recommendations. [Open source](#)
- S22** StatPearls. Implantable Intrathecal Drug Delivery System. [Open source](#)
- S23** Harden RN et al. Complex Regional Pain Syndrome: Practical Diagnostic and Treatment Guidelines, 5th edition. Pain Medicine, 2022. [Open source](#)
- S24** Shanthanna H et al. Evidence-based consensus guidelines on patient selection and trial stimulation for spinal cord stimulation, 2023. [Open source](#)
- S25** Troubleshooting intrathecal pumps in pain management: a clinical review. 2026. [Open source](#)
- S26** FDA. Important patient and consumer information about regenerative medicine therapies. US regulatory example; local rules differ. [Open source](#)
- S27** StatPearls. Opioid Toxicity. [Open source](#)
- S28** StatPearls. Diabetic Peripheral Neuropathy. [Open source](#)
- S29** StatPearls. Spinal Cord Stimulation. [Open source](#)
- S30** IASP. Definitions of Chronic Pain Syndromes. [Open source](#)
- S31** Narouze S et al. Interventional spine and pain procedures in patients on antiplatelet and anticoagulant medications, second edition. Regional Anesthesia and Pain Medicine, 2018. [Open source](#)
- S32** ASRA. Checklist for Treatment of Local Anesthetic Systemic Toxicity (LAST), 2020. [Open source](#)
- S33** Shirley Ryan AbilityLab. Rehabilitation Measures Database: Oswestry Disability Index. [Open source](#)